

ChronoMap 3D

Freezing Time for Open-Source Heritage

INFO90007 Social Computing – Assignment 2

The University of Melbourne

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1 Introduction

Three-dimensional models of physical spaces usually need dedicated equipment, like LiDAR scanners or purpose built 360 degree rigs which are expensive and impractical for most of the people to access. 3D data is available to the public but it usually comes from proprietary platforms like Google Street View. These platforms only cover the locations they choose and also updates them only when they decide to do so. There is no way for a member of the public or a local organisation to request a space be captured or re-captured more often. This creates a gap for spaces that change frequently, or even where up-to-date information matters, for example around the Royal Melbourne Hospital precinct where accessibility ramps, temporary walkways or the entrance layouts can shift with construction. Tourists, people planning a visit, checking accessibility, or navigating unfamiliar places are left relying on outdated two-dimensional references like OpenStreetMap which show layout but not depth. An openly accessible, reasonably current 3D capture would make it easier to navigate, plan and explore a space before even arriving there.

A 3D reconstruction can be thought of as a set of 2D photographs of the same space, taken from different angles and stitched together. This can be thought of as one person with one camera, who can capture a 3D space just by walking around it and taking photos from different positions. However, this will take a long time, often several minutes for a single location and the scene does not stay still during that time, lighting shifts as clouds move, shadows lengthen and background objects like parked cars or passing pedestrians appear in some photos but not others. When the scene changes between photos, stitching becomes much harder. Stitching photographs together assumes they all represent the same scene, ideally frozen at a single instant rather than captured gradually over several minutes. It is not just about matching visual features between images and stitching them but about deciding which photo, out of many taken at slightly different times, should count as the true reference for what the space looked like. A single, slow-moving capture process cannot solve this cleanly since every photo in the set was taken at a slightly different moment.

The system we propose, ChronoMap 3D, addresses this by treating a group of separate 2D cameras as parts of one distributed 3D camera. A group of participants position their devices around a target location and capture takes place for a short, defined time window. This means every photo in the set is taken close together in time, so the resulting collection is much closer to a single frozen instant than a sequence of moments recorded minutes apart. This is meant to reduce the inconsistency problems described above, though how narrow this time window can realistically be made and how much timing precision reconstruction actually needs are questions the later sections of this report come back to. The broader aim is for ChronoMap 3D to work as an open, community built source of current 3D spatial data as a way for the public to build and keep a shared record of the places that matter to them.

2 Method

2.1 Ideation & Concept Development Process

We arrived at the concept of ChronoMap 3D through an iterative brainstorming process, where team members shared ideas for potential areas of crowdsourced tasks. Initial ideas included capturing images for 3D reconstruction, transcription of live events, restaurant recommender systems and so on. These ideas were communicated both during in-person tutorials and during instant messaging (WhatsApp). To determine then evaluated the suitability and feasibility of each idea across the three main elements of the assignment requirements:

- **Crowdsourcing:** Does crowdsourcing of these tasks have a meaningful benefit?
- **Collocation:** Does the task require users to be in the same physical space?
- **Synchronicity:** Does the task require users to collaborate in real-time, or how does the task benefit from synchronous collaboration? How can the application facilitate and enhance this collaboration?

For example, we revisited feedback on the restaurant recommender concept. Participants could be in separate location to do the activity so it did not satisfy collocation. This helped clarify that connecting people through an application was not enough, they needed to share the physical setting.

We further evaluated the 3D reconstruction concept, with the idea that synchronous image capture could depict a moment "frozen in time"; however we also questioned the practical usefulness of this approach, given that historically 3D reconstructions successfully rely on asynchronously-captured or non-collaborative datasets. Ultimately through discussion we decided to go ahead with this concept based on its merits as discussed in Section 4.1.

We also considered framing the concept around capturing nature trail environments rather than landmarks and other architecture. However, we discarded this idea and pivoted back to the original concept, on the basis that (1) the synchronous element is not suitable capturing such a large scene (2) an online application could not rely on poor internet connectivity in remote areas.

Alongside these constraints, we discussed participant motivators and rewards, such as contribution recognition and shared souvenirs. We continued to develop and flesh out different features of the application, which are described in the following sections. The next steps were to draft paper sketches and review the design and heuristics together.

2.2 *Prototyping & Iteration Protocol*

We prototyped in three rounds, moving up in fidelity only once the previous round settled (Figures 1–6).

- **Hand-drawn paper sketches.** The first round was drawn by hand, covering the three scenarios that carried the most uncertainty: joining a session, recruiting passers-by on site, and creating a session. Paper was deliberate. It is quick to redraw, and a rough sketch invites the group to change it rather than defend it.
- **Scenario storyboard.** The second round placed those screens in context, following three named users through a visit to a landmark. This tested whether the collaboration made sense as a social situation, not only as a sequence of screens.
- **Low-fidelity wireframes.** The final round rebuilt the core screens in Figma, kept greyscale and unstyled so that review stayed on structure and wording rather than visual design.

Each round was reviewed together over Zoom against the community heuristics cards and the co-location and synchronicity requirements, and the issues raised fed directly into the next round.

2.3 *Community Heuristics Framework*

During the design process, we aimed to design the purpose and features of ChronoMap in a way that would identify and address suitable community heuristics for crowdsourcing platforms. We included the highlighted heuristics from the following categories:

- **Purpose:** Clarity, Visibility, **Idealism**
- **Moderation:** Monitoring, **Regulation**, Moderation
- **Members:** Self-presentation, Deep profiling, Life cycle, **Recruitment**
- **Common ground:** **Subgroups**, Diversity, **Events**
- **Contribution:** Comparative, **Request list**, **Targeted requests**, Threshold, **Motivators**, Request Description
- **Platform:** **Uniqueness**, Aesthetics, **Motive**, Rituals, Tools, Reputation, Virtual co-presence

These heuristics were considered while constructing the features as described in the following sections. For further discussion, refer to Section 4.2.

3 Results (Technical Architecture & Evolution)

3.1 Prototype Evolution

The first prototypes were paper sketches drawn for three scenarios: joining a capture session, recruiting passers-by with a QR code, and creating a new session (Figures 1–3). These were followed by sketches of the main screens (Figure 4). In these early sketches, a session was a shared time window, for example starting in 10 minutes and running for 5 minutes, and each participant pressed the shutter themselves as many times as they liked. The camera screen kept a standard manual shutter button.

Reviewing the sketches against the concept discussion in Section 2.1 exposed a gap. Photographs taken inside one window would be closer together in time, but nothing in the sketches told participants where to stand, and nothing showed the group how much of the site had already been covered. The Figma wireframes (Figure 6) respond to this in three ways. The camera screen keeps the manual shutter from the paper sketch and adds a session timer, so people photograph freely for the length of the window rather than waiting for a single trigger. A coverage guide was added alongside the camera, turning the loose idea of “covering different areas” from the first sketch into something the group can see. Discovery moved from typing a session code to a list of nearby sessions with a QR shortcut, while scheduled sessions remained an option. The share screen stayed largely the same, but it now credits every contributor as well as the host.

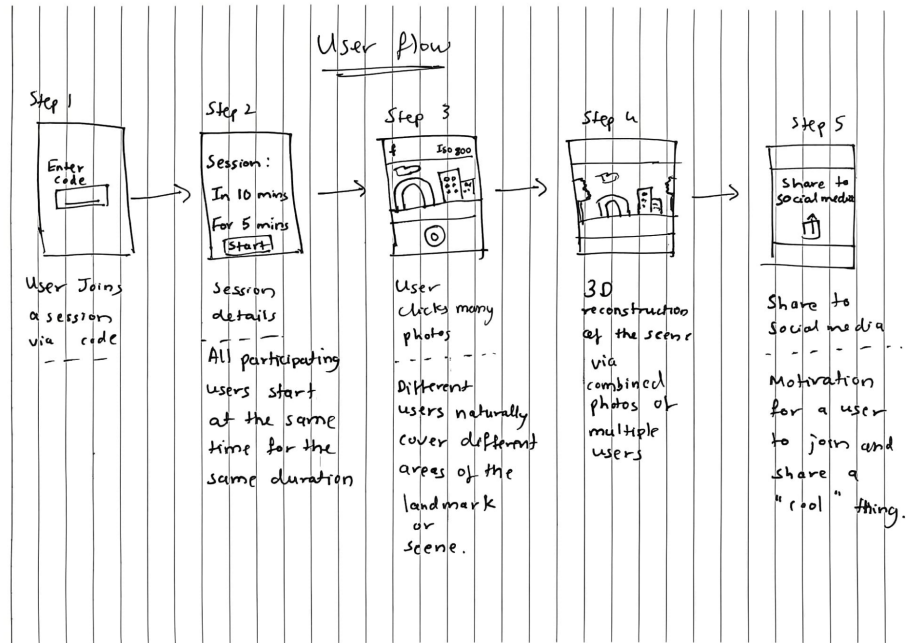


Figure 1. Low-fidelity paper sketch, Scenario 1: user flow for a group 3D capture. The user joins a session via a code, sees the shared start time and duration, takes photos, receives the 3D reconstruction built from everyone’s photos, and shares it to social media.

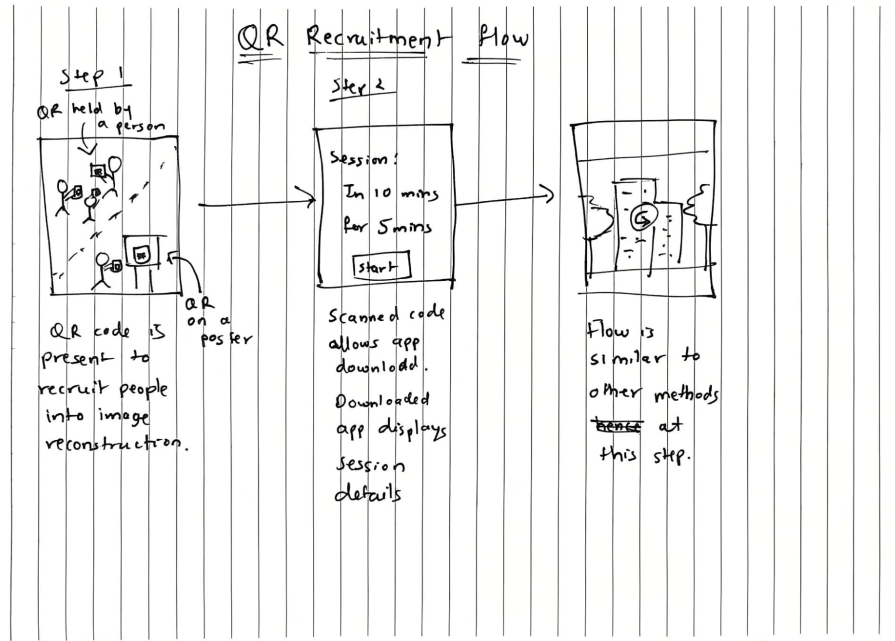


Figure 2. Scenario 2: instant recruitment using QR codes. A QR code held by a participant or printed on an on-site poster lets passers-by download the app and see the session details.

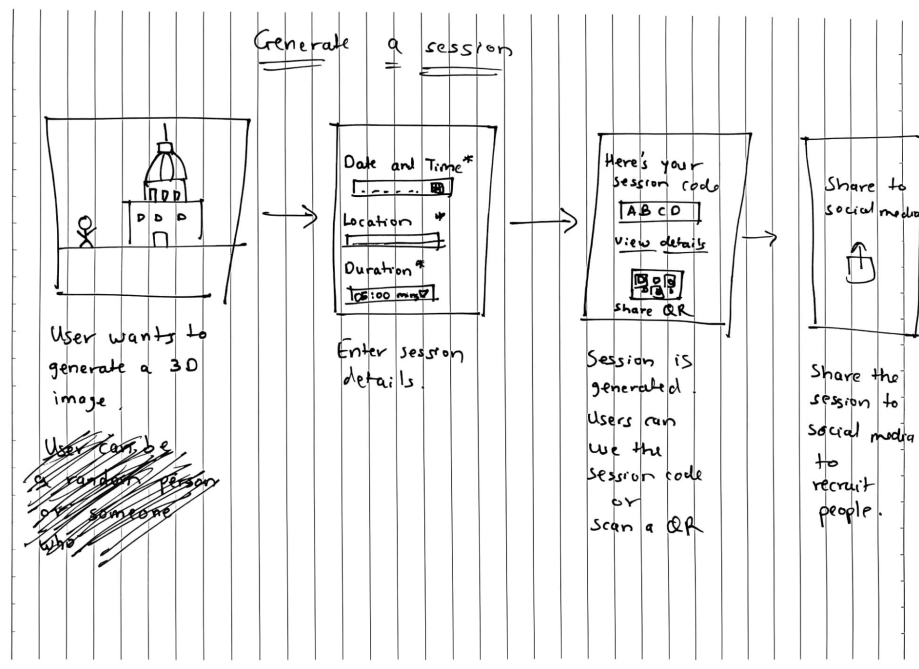
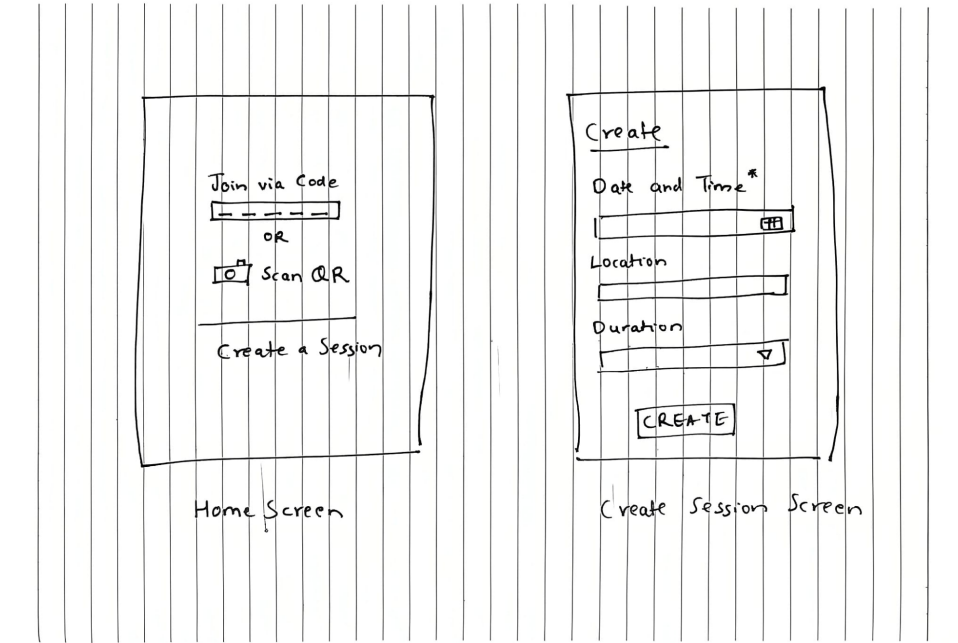
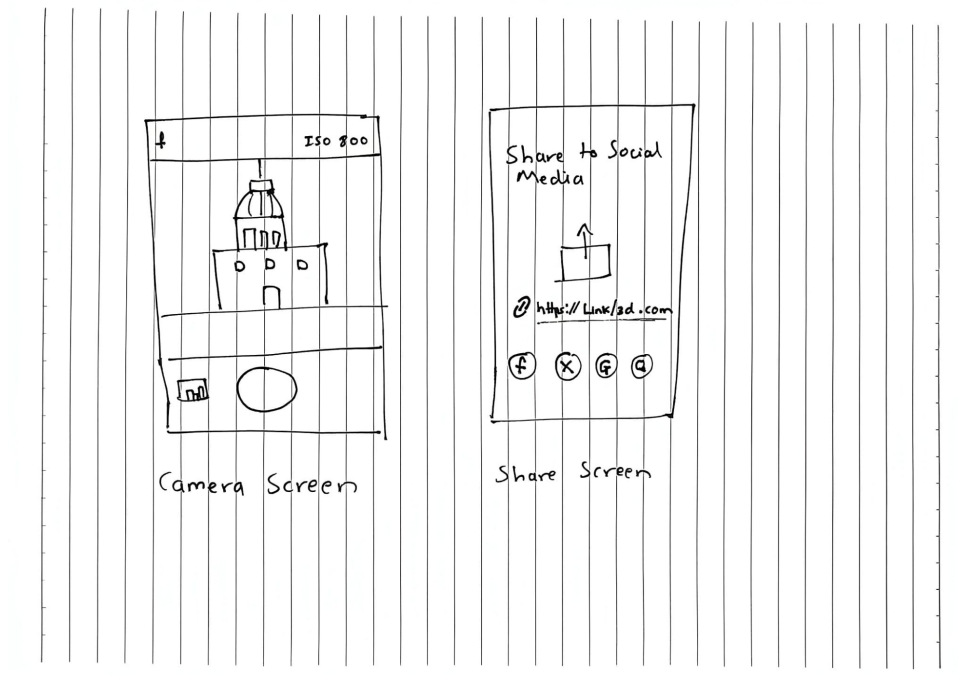


Figure 3. Scenario 3: a user generates a session by entering the date, time, location and duration, then shares the session code or QR to social media to recruit participants.



(a) Home screen and Create Session screen



(b) Camera screen and Share screen

Figure 4. Paper UI sketches of the four main screens.

3.2 End-to-End Scenario Storyboard

Iterating on these initial sketches, we created a scenario storyboard (Figure 5) illustrate a more fleshed-out scenario to demonstrate how users would use the application and complete the core tasks in context, highlighting how the application facilitates user collaboration in the collocated synchronous setting.

The storyboard depicts a scene where three friends, Alice, Bob and Charlie, are visiting a point of interest and see a poster advertising ChronoMap 3D (the recruitment stage). They decide to download the application, and upon startup the application shows them a tutorial of how ChronoMap 3D works,

making the purpose of the service visible and clear to the user. The application detects and identifies their current location, and prompts them to host or join an image capture session to photograph that location. Alice creates a session, which Bob and Charlie join via a shared 4-letter code.

When everyone is ready, Alice starts the session and the users take photos of the location from different angles. Alice, Bob and Charlie are able to talk to each other to co-ordinate where each person is taking their photos from and what angles have not been captured. The application also facilitates this coordination through detecting users' locations and comparing photographs taken so far, then prompting users to move to a different location, or photograph uncaptured gaps between themselves and other users.

At the end of the session, users are given a contribution score relative to each other, based on the number of photos and the quality of information captured. They are able to view and share their reconstruction at a later time after it has been processed by the system.

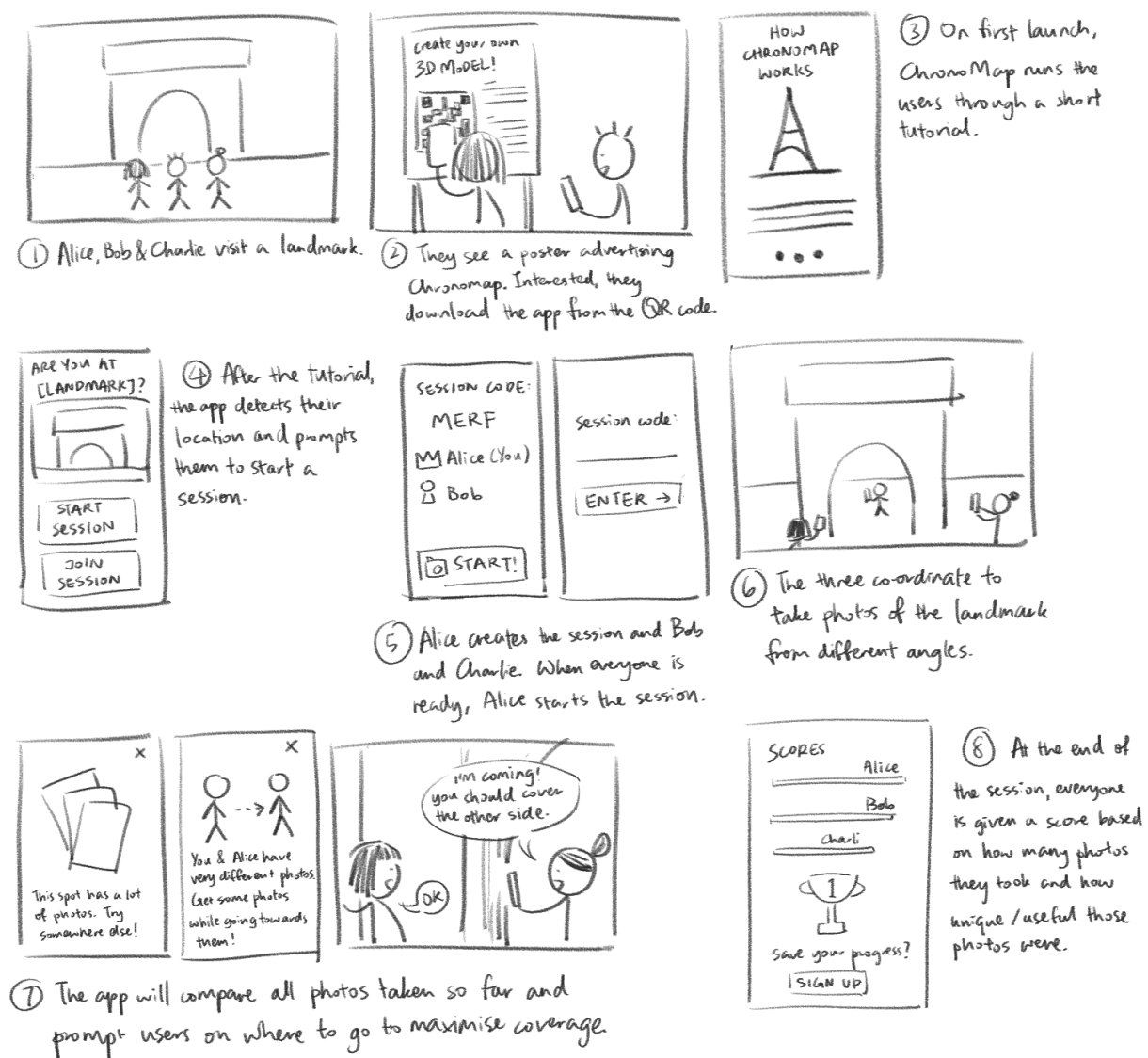


Figure 5. Scenario storyboard. Alice, Bob and Charlie find ChronoMap through a QR poster at a landmark, complete a short tutorial, form a session, coordinate capture angles using in-app coverage prompts, and receive contribution scores at the end.

3.3 Core System Interface & Technical Pipeline

Figure 6 shows the four core screens described below.

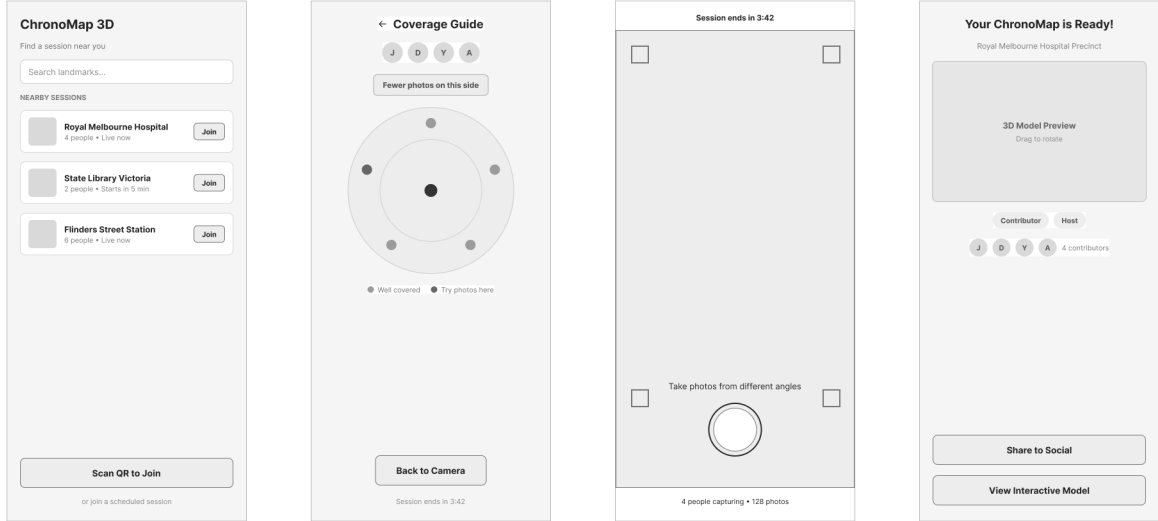


Figure 6. Low-fidelity Figma prototype of the four core screens: (1) session discovery and lobby, (2) coverage guide, (3) capture window with a manual capture button, and (4) result, souvenir and sharing.

3.3.1 Session Discovery & Group Formation

Users would find active sessions through nearby circles, scan an on-site QR code or join a pre-scheduled session. Each session would identify the target, purpose and participation status. For example, a session around the Royal Melbourne Hospital precinct could focus on documenting a publicly accessible exterior area. Joining would bring users into one shared lobby, where they could see who is present and what the group still needs before capturing.

3.3.2 Coverage Guide

While a session is running, the coverage guide shows roughly where photographs have been taken around the target, based on estimated participant positions and camera directions. Sides with fewer photographs are marked so that participants can move across and fill them in.

The guide offers suggestions rather than proof of complete coverage. Compass direction alone cannot establish whether an image contains enough overlap or whether an object blocks the view, and reconstruction requires overlapping images from different viewpoints¹. Section 4.3.3 returns to this as a limitation. In practice the guide supports the conversation already happening on site, where participants can tell each other which side looks thin.

3.3.3 Capture Window & Redundancy

Once the group is ready, the host starts a capture window of a few minutes. Participants photograph the target freely for that period, and the interface shows the time remaining rather than counting down to one shared instant. Coordination during the window is social. People move around the site, talk to each other and take different angles while the clock runs.

Collecting many photographs of the same surfaces is what handles movement in the scene. A pedestrian or a bird appears in only a few frames of any given area, so reconstruction can treat it as an outlier against the frames that do not contain it. The same redundancy helps where one participant’s view is blocked, because another person standing elsewhere has usually photographed the same surface.

Timing still matters. A longer window collects more photographs but allows shadows to shift and parked cars to leave, so window length trades scene consistency against coverage, and a workable length would need testing. After the window closes, images and available metadata are uploaded for quality checks and reconstruction. Participants see processing progress before receiving the model. The collocated and

synchronous part of the activity is the capture window itself, where people are in the same place at the same time and adjust to each other. Uploading and reconstruction happen afterwards.

3.3.4 Social Media Sharing & Contribution Rewards

After processing is complete, participants would receive a digital souvenir of the space they helped capture. The result screen would let them explore the reconstructed model and see their contribution recorded alongside the other participants. A contribution badge would recognise participation, while a separate host badge would acknowledge the person who organised the session.

From this screen, users could choose to share a preview image or a short rotating video of the model to social media, with a link to the interactive version. Similar to sharing a completed route on Strava, the post would show the shared outcome and let participants say, “I helped create this.” Users would preview the post and choose whether to include their contributor name before sharing.

These rewards respond to the waiting problem raised during brainstorming. The intention is to give participants something they can keep and share, while making their contribution visible. Whether this would motivate people to join or stay in a session would still need to be evaluated.

3.4 Gamification, Social Dynamics & Quality

A recurring concern raised during brainstorming was why a stranger would willingly coordinate with people they may not know while a group coordinates a synchronised photo.

The reward structure responds to this, a digital souvenir, every participant receives something tangible to keep regardless of their role in the session while the host is treated separately since organising a session takes extra coordination effort, the initiator is given a distinct incentive. Social sharing extends this visibility beyond the session itself, letting participants export their contribution to social media in the same way a shared running route on Strava turns a private activity into a visible one. Two mechanisms are proposed to sustain motivation. Locations could be captured as weather or seasonal sets, so a site is not considered “complete” until it has been documented across conditions such as spring, summer, golden hour or snow, giving contributors a reason to return to a familiar location rather than only seeking out new ones. Separately, participants visiting places that are difficult to synchronise, for instance requiring a specific time of day, weather condition or minimum group size rewards participants for solving a harder coordination problem.

4 Discussion

4.1 System Strengths & Collaborative Value

The main strength of the proposed system is that each participant contributes a viewpoint the others may not have. A single person moving around a site records different moments, while a coordinated group can aim to capture several perspectives at approximately the same time. This could reduce inconsistencies associated with movement between photographs. It should not be described as eliminating motion artifacts, because image quality, occlusion and shutter timing still affect the result.

The application also aims to make spatial data easier for the public to create and access. An open-access model could help people explore the visible layout of a public place, including entrances and ramps. Its value would depend on clear licensing, capture dates and quality information. A crowdsourced model would not, by itself, establish whether an access route is currently usable.

Co-location creates a further social benefit: participants can speak directly while coordinating. Someone could point out a blocked view, explain where coverage is missing or help another person understand the task. The radar supports this conversation by giving the group a shared view of progress. Together, these interactions make participation more than an individual upload. Each person can see how their actions contribute to a shared outcome.

4.2 Evaluation of Community Heuristics

Table 1 outlines explanations for how each identified community heuristic will be addressed by the system.

Category	Heuristic	Explanation
Purpose	Idealism	The platform's core aim is framed around digital archiving, local history and preservation.
Moderation	Regulation	The proposed interface would show rules and ethical guidelines, covering appropriate locations and privacy protections, before a session could be created or joined and again at application startup.
Members	Recruitment	Addresses how new contributors would actually be brought onto the platform, through shareable social media links and on-site QR codes displayed near a target location.
Common ground	Subgroups	Frequent contributors would likely cluster around shared interests over time, for instance around museums, gardens, public infrastructure or a specific city, rather than needing this structure imposed from the outset.
	Events	Allows users to share planned visit dates for a location, so that others intending to visit at a similar time could see this and arrange to meet, letting strangers coordinate a session together rather than relying on chance.
Contribution	Request list	The app will display a searchable list of locations to be captured, as well as a list of public, user-planned events. This allows users to discover planned group sessions local to them or relevant to their interests.
	Targeted requests	Given that the user grants location tracking permissions for the app to function, ChronoMap 3D can automatically detect when an existing user is at a point of interest and prompt them to host a session. This can be done when the user opens the application or by sending notifications.
	Motivators	Splits into extrinsic motivators, such as social media sharing and gamification, and intrinsic motivators, such as a sense of contributing to a community record or supporting local historians.
Platform	Uniqueness	A 3D capture retains a distinct novelty that a standard photograph does not, an effect the team expects to strengthen further.
	Motive	The application includes a landing tutorial that can be revisited, which explains the function and purpose of the system. It could also include an "about" which further explains the purpose and use of 3D reconstructions, and links to relevant organisations.

Table 1. Explanations for the chosen heuristics.

4.3 Technical Limitations & Edge Cases

4.3.1 Internet Connection

Internet connection is required to create, join, sync to a session, or to send photos to a server for them to get stitched into a 3D image. This requirement makes it difficult to use this app in remote areas where a lack of internet connection is a possibility.

4.3.2 Variation in Cameras

Different mobile devices have different kinds of camera modules in them and often even the same model of a phone can have a slight variation between lenses. This makes it a challenge to sync photos as photos can have different colour balance, focal length, center of lens, etc.

4.3.3 Uncaptured Areas after Final Processing

The boundaries of the 3D space are not defined initially and depend on the captured photos. Once the boundary of a 3D image has been calculated it could be possible that some portions of the 3D space inside that boundary have not been captured or have poor coverage with poor quality of photos. The coverage guide present in the application helps resolve some of this issue by guiding people towards less captured areas but does not consider the final output.

4.4 Spatial Safety, Ethics & Governance

The application should show safety warnings to encourage users to be mindful of their own safety (i.e., not wandering onto roads), and to not cause obstructions or violating privacy of strangers. Any private information, such as faces or license plates, should be addressed by the system by automatically detecting and blurring such objects, and if possible removing the object from the scene reconstruction entirely.

5 Conclusion

ChronoMap 3D provides a crowdsourced approach for 3D reconstruction of a scene. The system combines two ideas from two different fields. From multi-view photogrammetry, it explores the possibility of using multiple photographs of a single scene, ideally taken at the same moment. From social computing, it takes the idea that strangers who share a place can be organised into a group with a common goal. The proposed interface ties these together through co-located sessions, a shared coverage radar and a shared capture window. Further analysis is required to determine whether a shared capture window can provide enough detail for a 3D reconstruction, and whether souvenirs and badges are enough to motivate strangers to take time and collaborate on a task.

AI Use Statement

Yes. The group used Gemini (Google), ChatGPT and Codex (OpenAI) and Claude (Anthropic).

Gemini was used for brainstorming while we were generating and narrowing concepts. Claude was used to summarise the transcripts of our Zoom meetings into readable notes, and to turn our dot point notes into prose, including drafts of Sections 3.1, 3.3.2, 3.3.3 and 5 written from those notes and our prototypes. We reviewed and edited each draft before including it. ChatGPT, Codex and Claude were used for the coding side of the report, converting our shared Google Doc into LaTeX for Overleaf with its figures and captions, and revising two screens of the Figma wireframe.

The design decisions and the arguments are ours. These tools were used to save time on rewriting and formatting, not to generate the concept.

References

1. Schönberger JL, Frahm JM. Structure-from-motion revisited. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR); 2016 Jun 27–30; Las Vegas, NV, USA. p. 4104–13.